

Robotically Controlled Magnetic Laser Aiming for Surgical Applications

L. Burstein, A. Reyes, S. Chiang, G. Pittiglio, L. Fichera

Department of Robotics Engineering, Worcester Polytechnic Institute, USA
laburstein@wpi.edu

INTRODUCTION

Lasers have long played a central role in the surgical space, offering a contactless means to deliver therapeutic energy. Lasers can cut, coagulate, and ablate tissue. This versatility has made them part of the standard armamentarium across many procedures, ranging from ophthalmology to otolaryngology and neurosurgery [1].

Modern surgical laser systems can be broadly classified as either free beam or fiber-based [1]. Free beam systems use steering mirrors or galvanometric scanners to control laser aiming; these systems typically offer high precision and rapid beam positioning but require an unobstructed line of sight to the surgical site. Fiber-based delivery addresses this constraint by channeling laser light through a flexible optical fiber, facilitating integration with endoscopes and enabling access to anatomically confined or deep-seated regions not amenable to direct visualization. This improved accessibility, however, comes at the cost of aiming precision, as conventional fibers lack active beam steering at the distal tip.

Recognizing these limitations, several groups have recently proposed solutions to equip optical fibers with robotic distal steering mechanisms [2-5]. While effective, these systems rely on mechanical [2-4] or electro-magnetic [5] components for actuation, which are difficult to miniaturize beyond a certain point (among the reported prototypes, the smallest has a diameter of 6 mm [2], while others are on the order of 1 cm [3-5]), thereby posing a limit to the range of minimally-invasive procedures for which they are suitable.

In this paper, we introduce an alternative fiber steering strategy that leverages magnetic actuation to reduce the hardware footprint deployed inside the patient's body. The proposed approach, illustrated in Fig. 1, uses two permanent magnets: a small magnet mounted at the distal end of the optical fiber and a larger magnet that is robotically manipulated to induce controlled bending. In a clinical setting, the larger magnet would remain outside the patient's body, positioned sufficiently close to generate the required magnetic fields. This concept parallels the strategy described in [6], where fiber steering was achieved using a system of eight electromagnetic coils arranged around a working volume. By contrast, in the proposed method the external magnetic field is produced by a single permanent magnet, thereby simplifying actuation and minimizing the device footprint, while still offering sufficient degrees of freedom for fiber steering.

MATERIALS AND METHODS

To study the feasibility of the proposed fiber steering

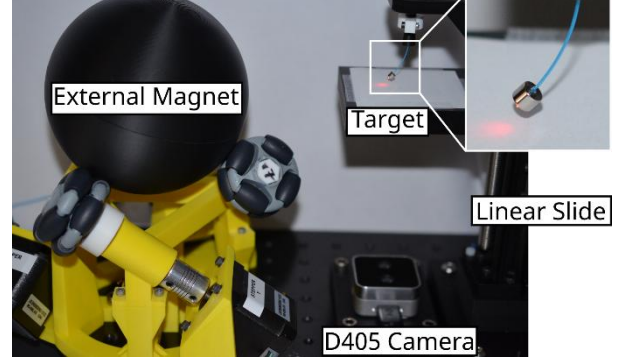


Fig. 1: Proposed approach for optical fiber steering. The external magnet is enclosed in a spherical 3D-printed shell, whose orientation is controlled by three omni wheels driven by stepper motors. Rotating the shell produces a change in the configuration of the magnetic field, which in turn alters the magnetic wrench applied at the optical fiber tip, where a smaller permanent magnet is mounted.

strategy, we assembled the experimental setup shown in Fig. 1: An off-the-shelf optical fiber, the FP200ERT (Thorlabs, NJ, USA) was outfitted with a cylindrical, hollow permanent magnet, the TUBE0077 (SM Magnetics, AL, USA), which has a diameter of 4 mm. A larger magnet, the DZ0X8-N52 (K&J Magnetics, PA, USA), was enclosed in a 3D printed sphere and positioned at a distance of 15 cm. Orientation of the sphere, and thus control of the magnetic field produced by the magnet, was achieved by means of omni wheels driven by stepper motors (4209L-01S, Lin Engineering, CA, USA), using the apparatus described in [7]. In the following sections, we first describe the kinematics model required to control the laser aiming, then report on benchtop laser tracking experiments demonstrating the fiber's ability to follow prescribed scan patterns.

Kinematic Modeling. Following the modeling approach of Kratchman *et al.* [8], we describe the deflection of the optical fiber using Kirchhoff Rod theory. Within this framework, the fiber shape is described as a space curve $\mathbf{p}(s)$ which satisfies the following differential relations:

$$\dot{\mathbf{p}}(s) = \mathbf{R}(s)\hat{\mathbf{e}}_3, \quad \hat{\mathbf{e}}_3 = [0 \ 0 \ 1]^T \quad (1)$$

$$\dot{\mathbf{R}}(s) = \mathbf{R}[\boldsymbol{\omega}_b]_{\times} \quad (2)$$

Here, $\mathbf{R}(s) \in \text{SO}(3)$ describes the orientation of the fiber at any point s along its length, while $\boldsymbol{\omega}_b$ represents the local curvature vector, with $[\boldsymbol{\omega}_b]_{\times}$ being its skew-symmetric matrix. Assuming linear-elastic material behavior with Young's modulus E and shear modulus G , the curvature $\boldsymbol{\omega}_b$ follows from the constitutive law

$$\boldsymbol{\omega}_b(s) = \mathbf{K}^{-1} \mathbf{R}^T(s) \mathbf{M}(s) \quad (3)$$

where $\mathbf{M}(s)$ is the internal bending moment and $\mathbf{K} = \text{diag}(EI, EI, GJ)$ is the stiffness matrix, with I and J denoting the second and polar moments of inertia of the

fiber, respectively. Assuming no distributed loading, the internal force $\mathbf{F}$ is uniform throughout the fiber and related to $\mathbf{M}(s)$ by the equilibrium expression

$$\dot{\mathbf{M}}(s) = -[\dot{\mathbf{p}}(s)]_{\times} \mathbf{F} \quad (4)$$

Numerical integration of (1-4) yields the fiber shape and the internal distribution of forces and moments. The force and torque applied at the fiber tip by the external magnet are imposed as boundary conditions at the free end $s = L$. These terms are computed by modeling both magnets as point dipoles with magnetizations $\mathbf{m}_i$ and $\mathbf{m}_e$, i.e.,

$$\mathbf{F}(L) = (\mathbf{m}_i \cdot \nabla) \mathbf{B} + \mathbf{G} \quad (5)$$

$$\mathbf{M}(L) = \mathbf{m}_i \times \mathbf{B} \quad (6)$$

$$\mathbf{B} = \frac{\mu_0}{4\pi\|\mathbf{r}\|^3} (3\hat{\mathbf{r}}\hat{\mathbf{r}}^T - \mathbf{I}) \mathbf{m}_e \quad (7)$$

with $\mathbf{G}$ denoting a gravity term, and $\mathbf{r}$ being the distance vector between the two magnets. In this work, the distal fiber magnet was installed in such a way that its dipole $\mathbf{m}_i$ was aligned with the fiber's longitudinal axis. The orientation of the dipole $\mathbf{m}_e$ was instead controlled by rotating the spherical shell (refer to Fig. 1).

To allow the execution of prescribed laser trajectories, the forward kinematics include an additional transformation that projects the laser beam onto the target surface, extending in a straight line from the fiber's tip. Inverse kinematics for the laser spot was solved with the Newton-Raphson method, using the central difference method to approximate the system's Jacobian matrix.

Experimental Procedures. We performed experiments to characterize the system's ability to track prescribed laser scanning paths. These experiments used a surgical KTP laser, the Aura i (Boston Scientific, MA, USA). The aiming beam of the laser was tracked with a D405 camera (Intel, CA, USA), using color-based image thresholding. Four scan patterns were tested, as illustrated in Fig. 2. To assess robustness to different spatial configurations, the vertical distance between the magnets was adjusted using a linear slider (Velmex, Inc., XN10). Each experimental condition was repeated five times, yielding a total of 60 trials. Tracking accuracy was quantified as the root-mean-square error (RMSE) between the commanded and measured laser paths.

RESULTS

The observed laser paths are compared to the reference paths in Fig. 2. The average RMSE was 1.02 mm (Standard Deviation: 0.36 mm) overall. Individual results for the scan shapes were: 0.75 mm (circle), 0.90 mm (infinity), 1.56 mm (sine wave), and 0.87 mm (square).

DISCUSSION

Experimental results show that the proposed magnetic fiber steering system was able to execute laser scans repeatably and with sub-millimeter accuracy across all but one experimental condition. An important limitation of this study is that the experiments were conducted in a benchtop environment, which does not capture clinically relevant challenges such as tissue motion. Additionally,

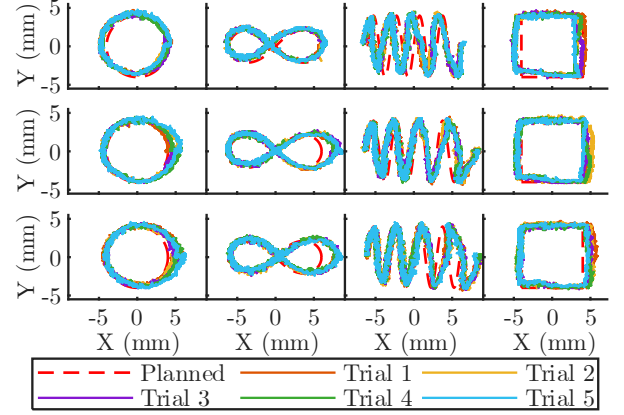


Fig. 2 Planned versus achieved laser paths. Each row represents a different external magnet height (Row 1: 10 mm, Row 2: 0 mm, Row 3: -10 mm).

all results were obtained in open loop, without compensation for modeling or actuation errors. Consequently, the performance reported here should be interpreted as a baseline for future work.

Combining flexible fiber optic laser delivery with the ability to achieve programmable, repeatable laser aiming has the potential to transform endoscopic laser procedures. Among these, we believe that one promising application area is the office-based treatment of lesions in the voice box, where current clinical workflows rely on hand-held fibers lacking any steering capability [9]. A steerable fiber capable of producing controlled rastering or scanning motions would amplify a physician's reach and increase the effectiveness of these procedures.

REFERENCES

- [1] Lee HC, Pacheco NE, Fichera L, Russo S. When the end effector is a laser: A review of robotics in laser surgery. *Advanced Intelligent Systems*. 2022 Oct;4(10):2200130.
- [2] York PA, Peña R, Kent D, Wood RJ. Microrobotic laser steering for minimally invasive surgery. *Science Robotics*. 2021 Jan 13;6(50):eabd5476.
- [3] Fang G, *et al.* Soft robotic manipulator for intraoperative MRI-guided transoral laser microsurgery. *Science Robotics*. 2021 Aug 18;6(57):eabg5575.
- [4] Lee HC, *et al.* Model-Free Control of a Soft-Rigid Hybrid Robot for Laser-Assisted Surgery. *IEEE Robotics and Automation Letters*. 2025 Nov 21.
- [5] Acemoglu A, Pucci D, Mattos LS. Design and control of a magnetic laser scanner for endoscopic microsurgery. *IEEE/ASME Transactions on Mechatronics*. 2019 Jan 30;24(2):527-37.
- [6] Charreyron SL, Gabbi E, Boehler Q, Becker M, Nelson BJ. A magnetically steered endolaser probe for automated panretinal photocoagulation. *IEEE Robotics and Automation Letters*. 2018 Dec 20;4(2):xviii-xiii.
- [7] Pittiglio G, *et al.* Magnetic Ball Chain Robots for Cardiac Arrhythmia Treatment. *IEEE Transactions on Medical Robotics and Bionics*. 2024 Sep 23.
- [8] Kratchman LB, *et al.* Guiding elastic rods with a robot-manipulated magnet for medical applications. *IEEE Transactions on Robotics*. 2016 Dec;33(1):227-33.
- [9] DeVore EK, *et al.* Benefits of Side-Firing Optical Fibers in Endoscopic Laser Treatment of the Larynx. *The Laryngoscope*. 2023 May;133(5):1205-10.